

Technical Data Sheet Epoxy Resin PX700K-1

Description

Epoxy resin PX700K-1 is a general purpose flame retardant potting and encapsulating compound. PX700K-1 has a long usable life and may be hot or cold cured. The system exhibits a good surface finish, high electrical strength, good thermal conductivity, low exotherm, and low cure shrinkage. The semi-flexible nature of the system allows thermal expansion without cured stress. PX700K-1 is compatible with most circuit board components and materials over an extended temperature range. The flame-retardants in PX700K-1 are of a non-halogen type and do not contain heavy metals. Adhesion is excellent to most plastics and substrates. It is available in bulk, kits, and twinpack form. The standard colour is black but other colours are available on request.

Features

High electrical insulating characteristics
Good thermal conductivity
Low shrinkage
High adhesion
Good chemical and water resistance
Meet the requirements of WEEE and RoHS

Specification

Property	Resin RX700K-1	Hardener HX700K-1	Mixed PX700K-1
Colour	Black	Amber	Black
Specific Gravity g/ml	1.8	1.00	1.7
Viscosity m.Pa.s @ 25°C	80000	350	12500
Mix ratio by weight	9.4:1		
Mix ratio by volume	5.3:1		
Usable life (150g @ 25°C)	120 minutes		
Gel time (150g @ 25°C)	360 minutes		

Approvals

RoHS compliant	Yes
UL94-V0	Yes
REACH (SVHC concentration)	Compliant

Cure Schedule

Minimum cure	Full cure
24 hrs @ 20°C	1 week
2 hrs @ 60°C	4 hours
1 hr @ 80°C	2 hours

The above are typical values and will vary depending on the cured mass and application. Hotter temperatures may be used for faster cure but will result in higher post cure shrinkage and higher cure exotherm. Experimentation and testing is suggested to avoid side effects. For maximum properties a post cure may be required - call Robnor Technical Service Department for advice.

Typical Properties

Peak Exotherm (150g @ 25°C)	40
Shrinkage % (Volume)	0.3
Thermal conductivity	1.00 W/mK
Operating temperature range*	-40 to +150°C - application & geometry dependent
Dielectric strength	18 kV/mm
Volume Resistivity	10 ¹² ohm x cm
Shore D hardness	80
Flame Retardancy	Approvable
Tensile strength	50 mPa
Compressive strength	60 mPa
Deflection temperature	50°C
Co-efficient of expansion	40 - 50 ppm/°C
Loss Tangent	0.060 @ 50 Hz
Permittivity	4.8 @ 50 Hz
Continuous tracking index	>850V (Method IEC 60112)
Water absorption	0.7% (30 days @ 20°C)
Elongation at break	2-5%
Oxygen index	38
Flexural strength	Semi-flexible

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Buy On-line: www.resins-online.com

Packaging

PX700K-1 is available in Bulk, Twinpacks & kits

Availability:

Available through distribution and www.resins-online.com

Twinpacks

Twinpacks are pre-weighed resin and hardener components contained in a tough flexible film, separated by a removable clip and rail. Once the clip and rail is removed, the resin and hardener can be thoroughly mixed within the bag and is then ready for use. Mixing will normally take ~ 3 minutes depending on the operator and viscosity of the material. Twinpacks are ideal for small to medium production runs, prototyping, and on-site or field use. The twinpack weight/volume may also be tailored to a specific size on request.

For further details, please visit www.robnor.co.uk

Bulk Material

PX700K-1 is a filled system and formulated to avoid sedimentation. However, if sediment is found after storage, this must be re-dispersed in the original container before use. Failure to do so may result in defective product. Long-term sedimentation will be aggravated by storage above 25°C and should be avoided. In bulk or kit form gentle mixing with a paddle or spatula will homogenise the material.

In bulk or kit form, evacuation may be necessary for best results.

Kits

In kit form, resin and hardener are provided in separate containers to the correct ratio.

In most cases, pour the hardener into the larger resin container and use it as a mixing vessel.

Stir well using an appropriate mixer until homogeneous.

Note: Incomplete mixing will be characterised by variable/partial cure (even after extended time periods).

Cleaning

All equipment contaminated with mixed material should be cleaned before the material has hardened.

Robnor Resins TS130 is suitable non-flammable cleaning agent, although other solvents may be found suitable. TS130 will also remove cured material provided it is allowed to soak for a number of hours.

Storage and Shelf Life

Material stored in the original unopened containers in cool dry condition between 10 and 25°C will have a shelf life of at least two years.

Once used the containers must be kept sealed to prevent effects from water, air, or contaminants.

Health and Safety

Epoxy resin systems may cause sensitisation by skin contact or inhalation may be corrosive, harmful, or toxic. It is therefore strongly recommended that skin and eye contact is avoided by the using of appropriate personal protective equipment such as gloves, safety glasses or goggles and overalls.

Wash any contamination from the skin immediately and thoroughly and do not eat, smoke or drink in the working vicinity. Under normal working conditions a good source of ventilation is adequate, however if the material is heated then local exhaust ventilation (LEV) may be required especially for curing ovens.

The above is given as a guide only; please refer to RX/HX700K-1 Health and Safety data or our Technical Service Department for individual/specific advice.

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